Q. Why atoms combined together?

⇒ 1) Due to intermolecular force of attraction

2) To archive the electronic configuration of nearest ~~inert~~ inert gas.

3) To follow the octet rule

4) To gain stable electronic configuration

5) To form new molecular.

Q. How does combine together?

⇒ 1) Transfer of electrons between two atoms (Ionic or electrovalent bond)

2) Equal sharing of electrons between two atoms (Covalent bond)

3) Sharing of electrons between two atoms by an atom. (Co-ordinate bond)

Q. Define chemical bond and types of it?

⇒ ~~The~~ Chemical bond:- The intermolecular force of attraction due to which two or more atoms combine together to form a new molecule is called chemical bond.

Chemical bond are mainly three types:

1) Ionic bond:| Transfer of electrons between two atoms where one atom is electropositive and other atom is electro-negative. Examp:| NaCl, KCl etc

2) Covalent bond:| Equal sharing of eletrons between two atoms where both ~~electro~~ atoms are electronegative.

Examp:| H_2O, SO_3 etc

3) Co-ordinate bond:| The bond formed when one sided sharing of electrons take place is called Co-ordinate bond. Examp:| NH_4^+, Al_2Cl_6 etc

4. Define Hybridization.

⇨ Hybridization:| When two or more orbitals combine together then a new types of orbital is forward the orbitals are called Hybrid orbitals and the process is called hybridization.

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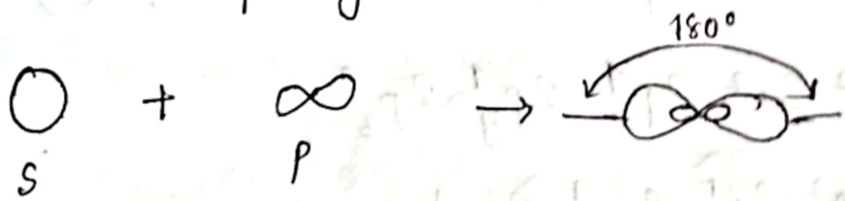
[4]. Discuss the bond formation process with bond angle, structure, s-character and p-character of $BeCl_2$, ~~BeF_3~~ BCl_3 and CCl_4 molecules through hybridization.

⇨

Sp hybridization: This hybridization involve the mixing of one s and one p orbital resulting in the formation of two equivalent sp hybrid orbitals.



$BeCl_2 \rightarrow$

$Be(4) \rightarrow 1s^2\ 2s^2$

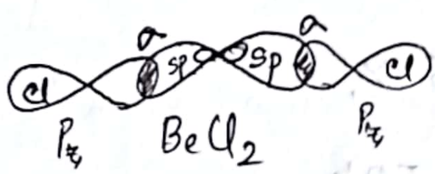
$Be^*(4) \rightarrow 1s^2\ 2s^2\ 2p_x^0\ 2p_y^0\ 2p_z^0$

$Be^{**}(4) \rightarrow 1s^2\ 2s^1\ 2p_x^1\ 2p_y^0\ 2p_z^0$

$Cl(17) \rightarrow 1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^5$

$Cl^*(17) \rightarrow 1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p_x^2\ 3p_y^2\ 3p_z^1$

~~diagram of~~ **diagram / structure:**



which is linear.

Bond Angle: 180°

S-character: 50%

p-character: 50%

~~sp~~

Extra: for C_2H_2 or $(CH \equiv CH)$ → 2π, 3σ total Bond · $2+3=5$

$C(6) \rightarrow 1s^2\ 2s^2\ 2p^2$

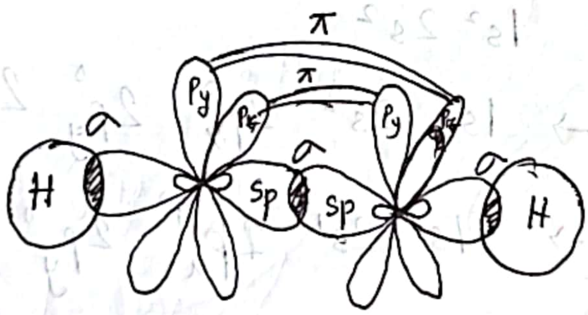
$H(1) \rightarrow 1s^1$

$C^*(6) \rightarrow 1s^2\ 2s^2\ 2p_x^1\ 2p_y^1\ 2p_z^0$

$C^{**}(6) \rightarrow 1s^2\ 2s^1\ 2p_x^1\ 2p_y^1\ 2p_z^1$

$\equiv \Rightarrow$ 2π, 1σ

$= \rightarrow$ 1π, 1σ



$CH \equiv CH$

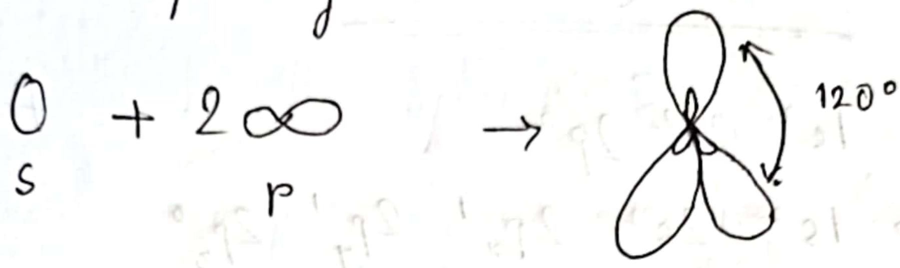
Structure: Linear

Bond Angle: 180°

S-character: 50%

p-character: 50%

sp^2 hybridization: In this hybridization there is involve of one s and two p orbitals in order to form three equivalent sp^2 hybrid orbitals.



$BCl_3 \rightarrow$

~~B(5) BCl~~$_3$

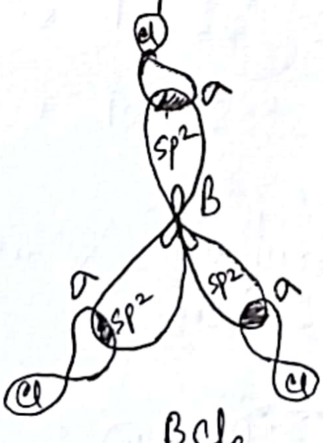
$B(5) \rightarrow 1s^2\ 2s^2\ 2p^1$

$B^*(5) \rightarrow 1s^2\ 2s^2\ 2p_x^1\ 2p_y^0\ 2p_z^0$

$B^{**}(5) \rightarrow 1s^2\ 2s^1\ 2p_x^1\ 2p_y^1\ 2p_z^0$

$Cl(17) \rightarrow 1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^5$

$Cl^*(17) \rightarrow 1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p_x^2\ 3p_y^2\ 3p_z^1$



BCl_3

Structure: Triangular

Bond Angle: 120°

s-character: 33.33%

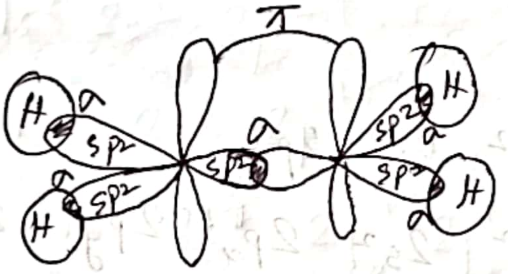
p-character: 66.66%

Extra: C_2H_4 or $(CH_2=CH_2)$

$C(6) \rightarrow 1s^2\ 2s^2\ 2p^2$ $H(1) \rightarrow 1s^1$

$C^*(6) \rightarrow 1s^2\ 2s^2\ 2p_x^1\ 2p_y^1\ 2p_z^0$

$C^*(6) \rightarrow 1s^2\ 2s^1\ 2p_x^1\ 2p_y^1\ 2p_z^1$



C_2H_4

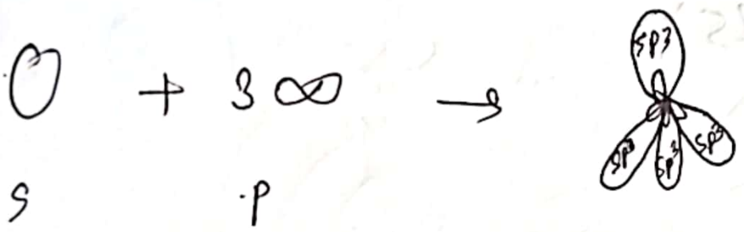
Structure: Triangular

Bond Angle: 120°

s-character: 33.33%

p-character: 66.66%

sp³ hybridization :- In hybridization, there is involve of one s and three p orbitals in order to form their equivalent sp³ ~~hybrids~~ hybridized orbital.



$CCl_4 \longrightarrow$

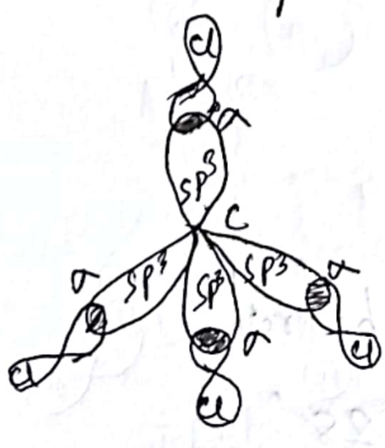
$C\ (6) \rightarrow 1s^2\ 2s^2\ 2p^2$

$C^*\ (6) \rightarrow 1s^2\ 2s^2\ 2p_x^1\ 2p_y^1\ 2p_z^0$

$C^{**}\ (6) \rightarrow 1s^2\ 2s^1\ 2p_x^1\ 2p_y^1\ 2p_z^1$

$Cl\ (17) \rightarrow 1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^5$

$Cl^*\ (17) \rightarrow 1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p_x^2\ 3p_y^2\ 3p_z^1$



CCl_4

Structure: Tetrahedral

Bond Angle: $109^\circ 28'$

s-character: 25%

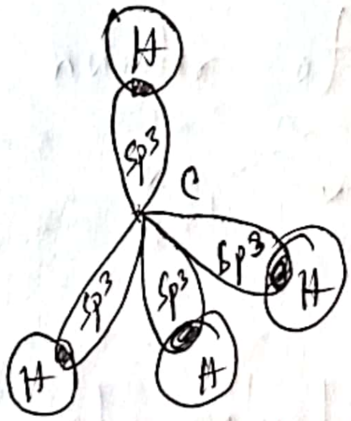
p-character: 75%

Extra: CH_4

$C(6) \rightarrow 1s^2\ 2s^2\ 2p^2$ $\qquad$ $H(1) \rightarrow 1s^1$

$C^*(6) \rightarrow 1s^2\ 2s^2\ 2p_x^1\ 2p_y^1\ 2p_z^0$

$C^{**}(6) \rightarrow 1s^2\ 2s^1\ 2p_x^1\ 2p_y^1\ 2p_z^1$



CH_4

Structure: Tetrahedral

Bond Angle: $109^\circ 28'$

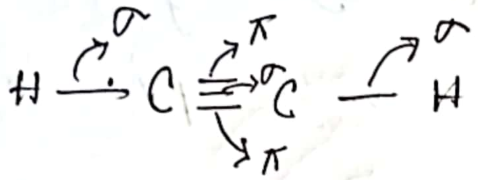
s-character: 25%

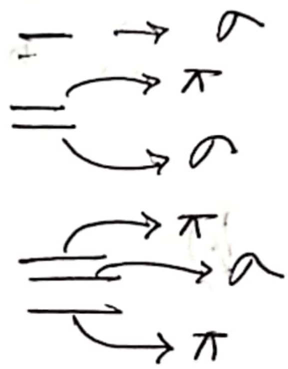
p-character: 75%

Q. What is the total number of sigma and pi bonds in the following molecules?

i) C_2H_2 ii) C_2H_4 iii) H_2O iv) SO_3 v) SF_6

⇒ i)

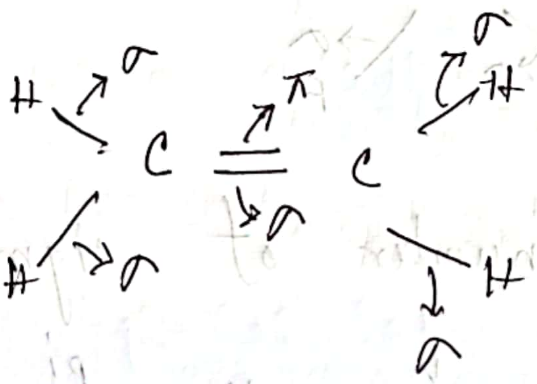




∴ the total number of sigma bond = 3

∴ " " " " pi " = 2 (Ans)

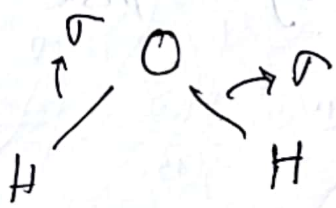
ii)



∴ the total number of sigma bond = 5

∴ " " " " pi " = 1 (Ans)

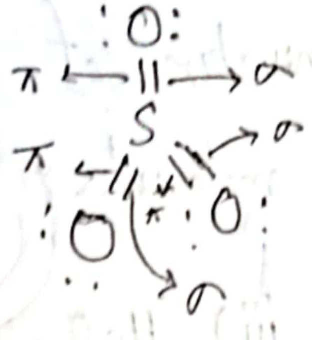
iii)



∴ the total number of sigma bond = 2

∴ " " " " pi " = 0 (Ans)

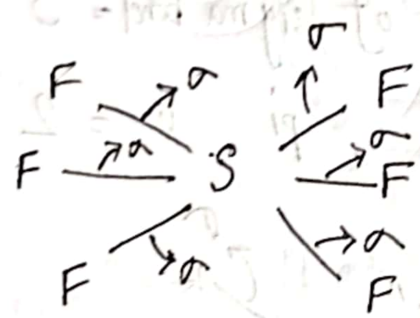
iv)



∴ the total number of sigma bond = 3

∴ " " " " pi " = 3 (Ans)

v)



∴ the total number of sigma bond = 6

∴ " " " " pi " = 0 (Ans)

4. Analyze and show clearly - the bond formation and orbital diagram of these molecules : NaCl and HCl

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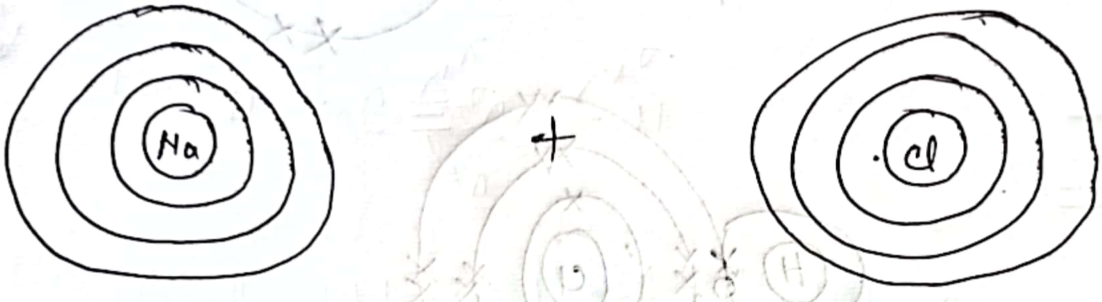
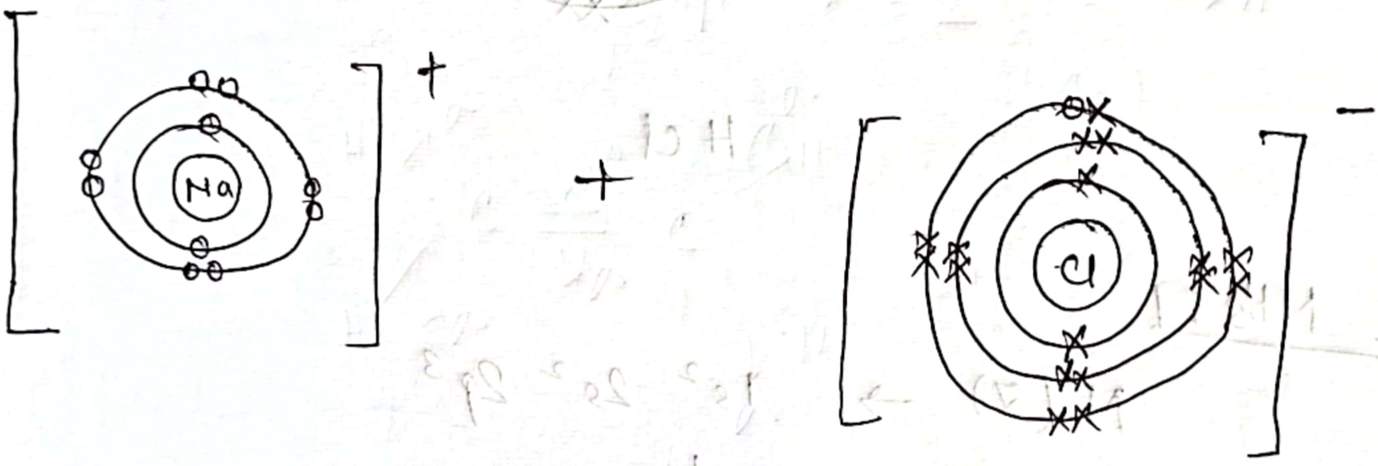
⇨ Ionic bond :

Na (11) → $1s^2\ 2s^2\ 2p^6\ 3s^1$

Cl (17) → $1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^5$





NaCl

Covalent bond :

H (1) → $1s^1$

Cl (17) → $1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^5$

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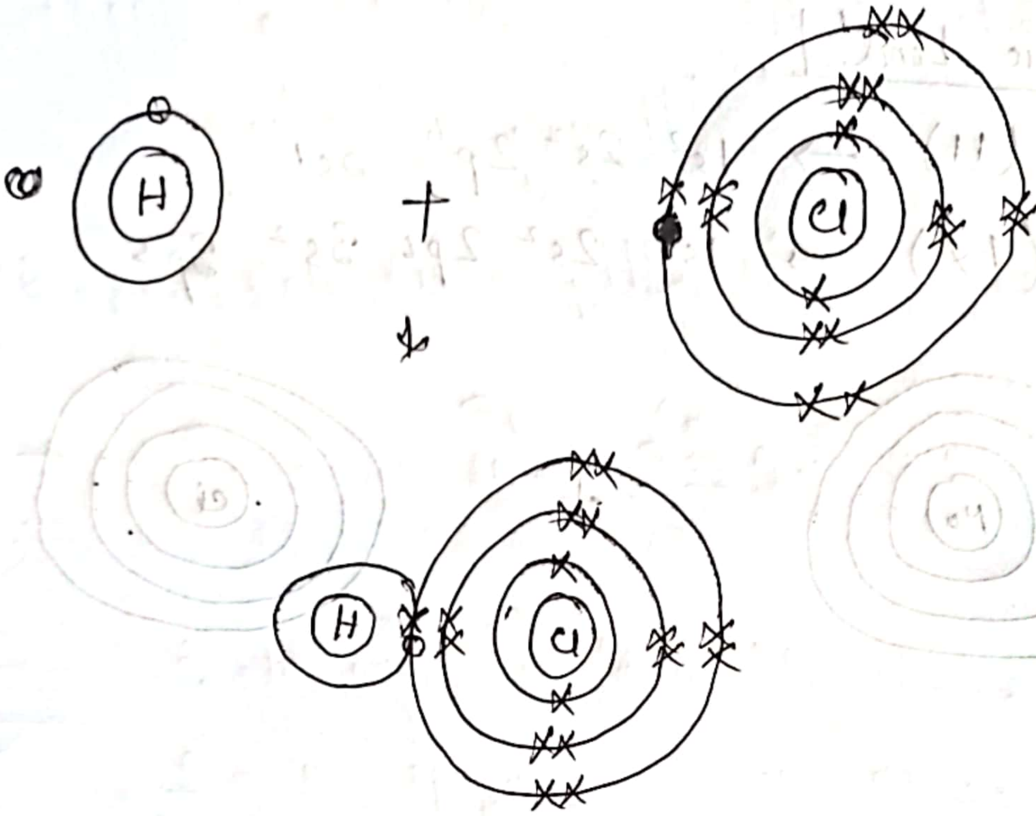
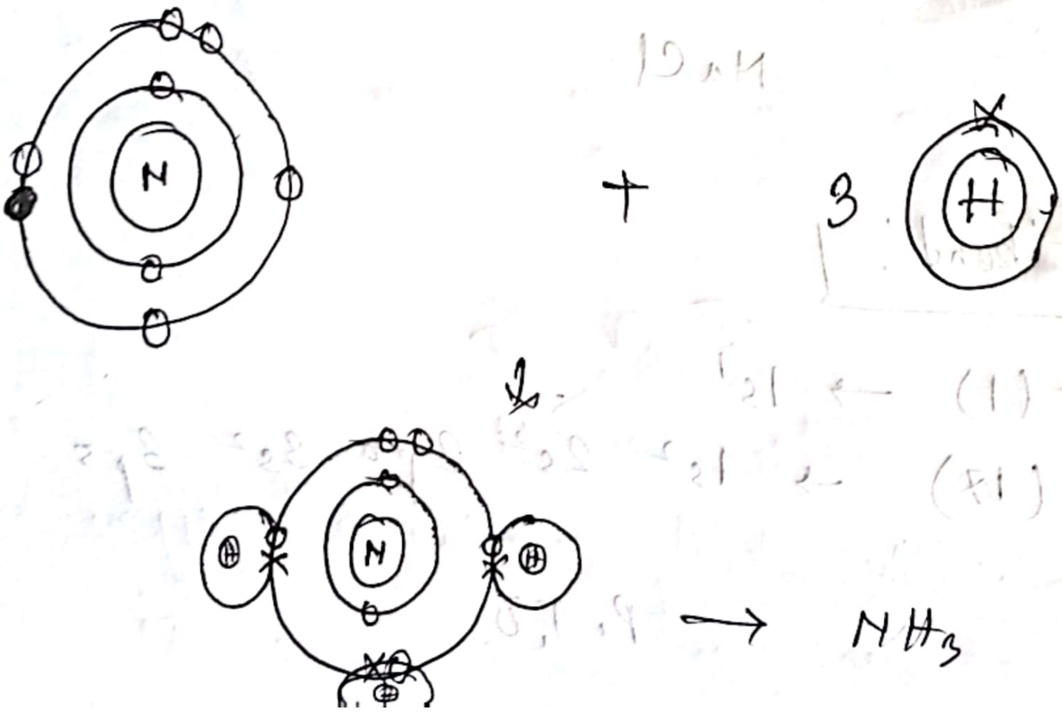


H Cl

NH$_3$:

N (7) $\rightarrow$ $1s^2\ 2s^2\ 2p^3$

H (1) $\rightarrow$ $1s^1$



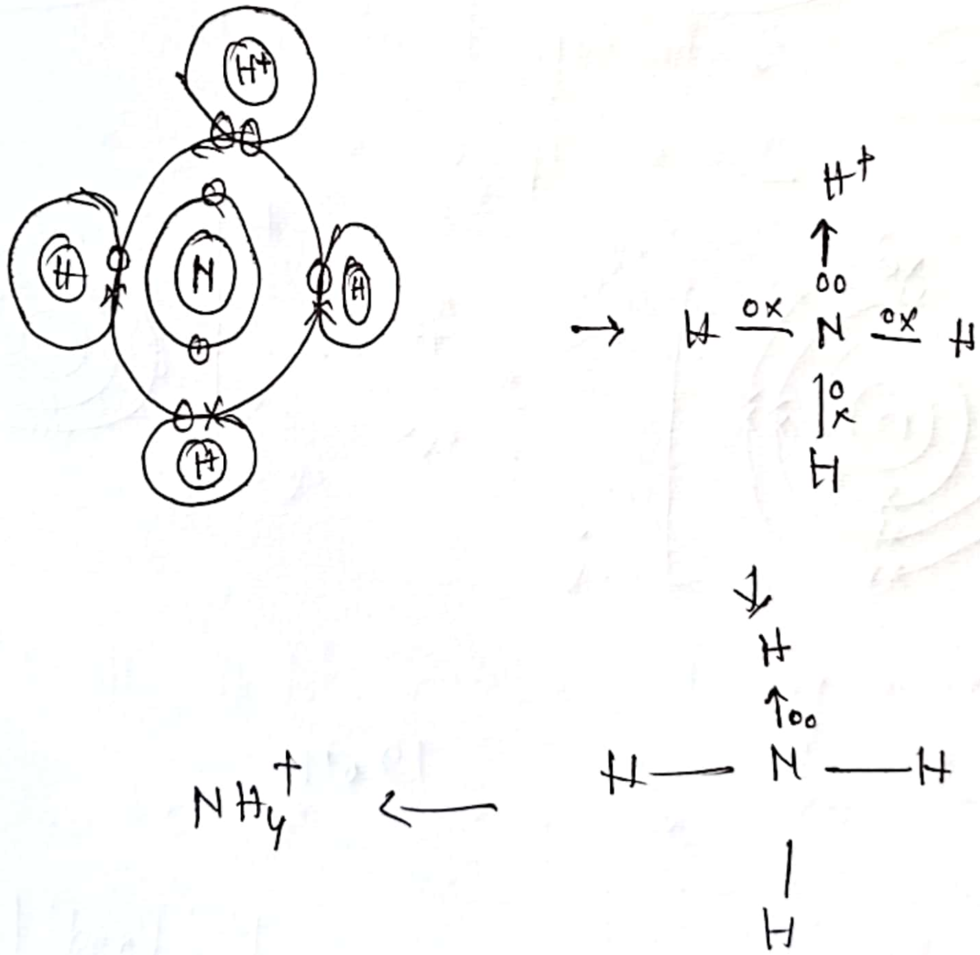
Co-ordinate bond:

$N\ (7) \rightarrow 1s^2\ 2s^2\ 2p^3$

$H\ (1) \rightarrow 1s^1$

$NH_4^+ \rightarrow NH_3 + H^+$

$\therefore H^+ (0) \rightarrow 1s^0$



Hybridization	Structure	Angle bond
sp	Linear	180°
sp^2	Triangular	120°
sp^3	Tetrahedral	109.5° / 109° 28'
sp^3d	Triangular Bipyramid	120°, 90°
sp^3d^2	Octahedral	90°
sp^3d^3	Pentagonal bipyramid	72°, 90°